

Privacy Attacks and Countermeasures in Biometric Gait Identification

Master Degree in Cybersecurity | Academic Year 2025/2026

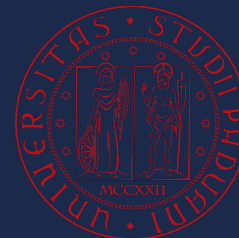
Tommaso Danieli

Student ID: 2152871

Supervisor: Prof. Simone Milani · Co-Supervisor: Mattia Tamiazzo

University of Padova — Master Degree in Cybersecurity

23 September 2026



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

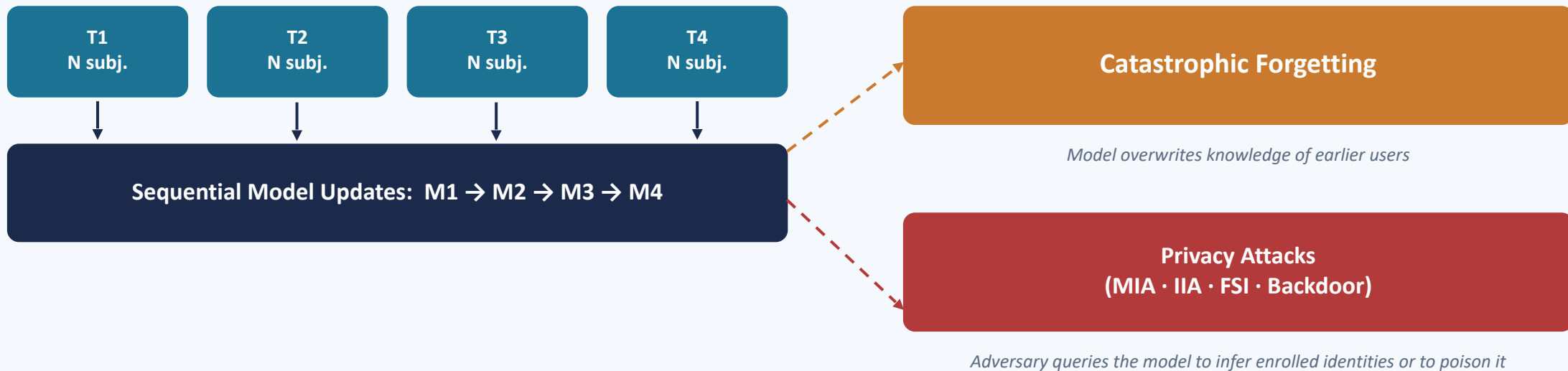


DIPARTIMENTO
MATEMATICA

The Problem: Continual Learning Meets Privacy

Gait recognition from wearable IMUs (*Inertial Measurement Unit*) authenticates users continuously, without active cooperation. But real systems enroll new users over time, thus the model must adapt without erasing old identities or leaking private information.

New subject cohorts arrive sequentially:



Evaluated via two standard metrics — average accuracy ($\bar{A}$) and backward transfer (BWT), which quantifies forgetting.

Three Datasets, One Approach

All three datasets share the same six-channel IMU structure and the same four-task continual learning split approach.

Property	WU Gait	UCI HAR	WISDM
Subjects	118	30	51
Channels	6	6	6
Window size (samples)	128	128	200
Sampling rate	50 Hz	50 Hz	20 Hz
Window duration	2.56 s	2.56 s	10.0 s
Task splits (T1–T4)	30 / 30 / 30 / 28	8 / 9 / 7 / 6	13 / 13 / 13 / 12

Spans 30–118 subjects, 20–50 Hz, and 128–200-sample windows — testing generality across different deployment conditions.

Four Privacy Attacks Against the Models

MIA

Membership Inference (MIA)

Black-box, per-sample — was window x used in training?

score = $-L_{CE}(f(x), y)$ — exploits the train/test loss gap

IIA

Identity Inference (IIA)

Black-box, per-subject — was person s ever enrolled?

Aggregates $N=20$ query windows per subject; mean CE score thresholded at τ

FSI

Feature Space Inference (FSI)

White-box — k -NN probe directly on the embedding layer

k -NN ($k=5$) on raw backbone embeddings h — bypasses the classifier head entirely

BD

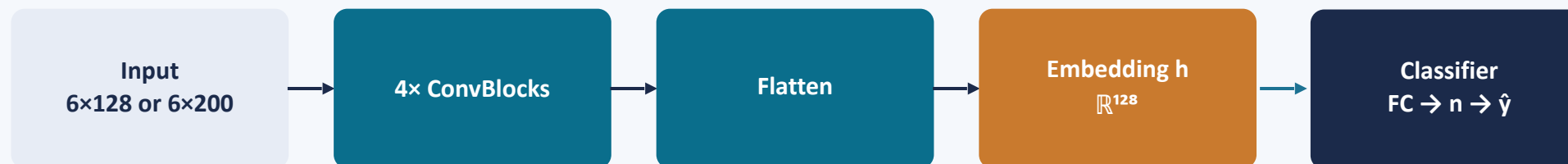
Backdoor Attack

Training-time — poisons data to implant a trigger $\rightarrow$ target mapping

Sinusoidal trigger on 15% of Task 3 windows, relabeled to the target subject of Task 4

Each attack is evaluated under two attacker modes: a realistic no-seed / no-adaptor attacker, and a worst-case oracle who knows the secret key.

GaitCNN: The Shared Backbone



Single-task upper bound (100 training epochs, no continual constraint):

92.1%

WU Gait · chance 0.85%

95.1%

UCI HAR · chance 3.33%

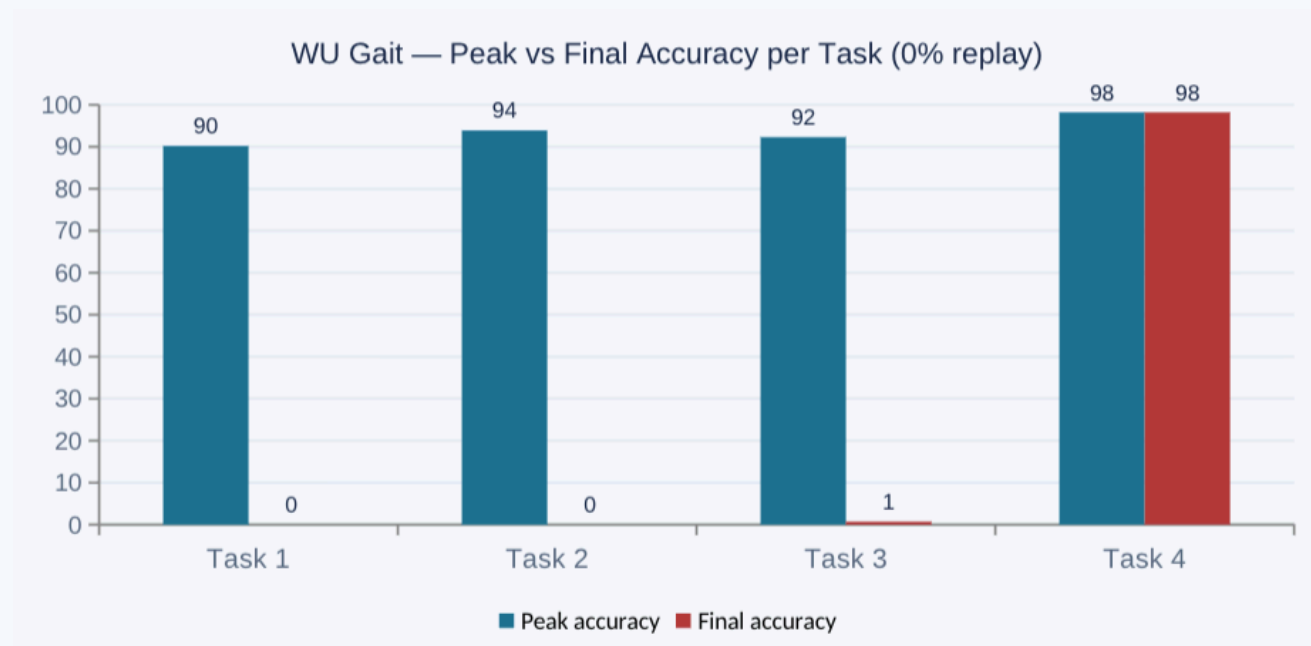
96.6%

WISDM · chance 1.96%

Confirms sufficient model capacity on all three datasets before the continual-learning constraint is introduced.

Catastrophic Forgetting: Why Naïve Fine-Tuning Fails

Under standard sequential fine-tuning with no protection mechanism, accuracy on each task collapses almost to zero as soon as the next task is learned.



3 Approaches to mitigate Catastrophic Forgetting

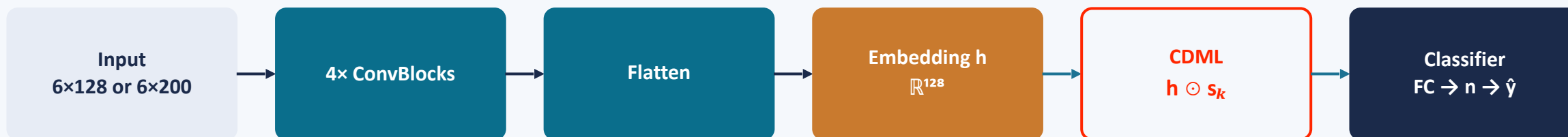
Regularization - Penalize drift on important weights
(*Knowledge Distillation, LiDER*)

Replay - Buffer or generative rehearsal of past tasks
(*Standard with 15% Replay, Wavelet Generative Replay*)

Architectural - Dedicated per-task parameters
(*CDML, LoRA*)

Backward transfer (BWT): -90.2 pp (Task 1) | -93.9 pp (Task 2) | -91.6 pp (Task 3) | Task 4 unaffected, as it is the last task trained.

CDML Cuts Forgetting at the price of 4 Bytes per Task



APPROACH

The **embedding** is multiplied by a **task-specific** ± 1 sequence, $m = s_k \odot h$. Sequences are near-orthogonal (cosine similarity ≈ 0), so tasks occupy separate regions of the embedding space.



t-SNE (*t*-distributed Stochastic Neighbor Embedding) projections confirm CDML spreads the four task clusters more uniformly across the embedding space than Std. This is a visual evidence of the orthogonal modulation at work.

Method	Peak acc. T1	Final acc. T1	Forgetting (Δ)	Bytes / task
Std (0% replay)	90.2%	0.0%	-90.2 pp	0 B
CDML (0% replay)	89.8%	51.0%	-38.8 pp	4 B

Privacy Protection Collapses Without the Seed

Across all four attacks, the no-seed attacker performs near chance, however an oracle who knows the seed recovers full identity information: CDML's protection is **access-control**, not representation obfuscation.

Attack	Metric	Std	CDML no-seed	CDML oracle
MIA	AUC	56.9%	51.0%	56.0%
IIA	AUC	66.4%	48.7%	99.2%
FSI	k-NN acc.	87.7%	1.6%	93.6%
Backdoor	ASR	10.5%	3.2%	64.7%*

*"Oracle" for Backdoor = tested with the poisoned task's own sequence (Task 3, seed 1002); any other sequence gives $\leq 8\%$ ASR.

Sequence-Dependent Backdoor

Only the poisoned task's own seed reactivates the trigger, a compromised backdoor stays contained to the users holding that one sequence.

Seed-Identification Risk

Given the list of 4 seeds, an attacker recovers the most recent task's seed 95.2% of the time by probing for max confidence, thus the seed list itself must stay secret.

Three Extensions to Close CDML's Forgetting Gap

KD

CDML + Knowledge Distillation

Focal distillation anchors the student to the teacher on samples it already got right -> targets negative flips

$$\mathcal{L}_{\text{total}}(\mathbf{x}, y) = \mathcal{L}_{\text{CE}}(f_{\theta}^{(k)}(\mathbf{x}), y) + \lambda \cdot (\alpha + \beta \cdot \mathbf{1}[\hat{y}_{\text{old}} = y]) \cdot \|f_{\theta}^{(k)}(\mathbf{x}) - f_{\theta}^{(k-1)}(\mathbf{x})\|_2^2$$

WGR

CDML + Wavelet Generative Replay

Synthesizes windows from stored statistics -> no raw data retained

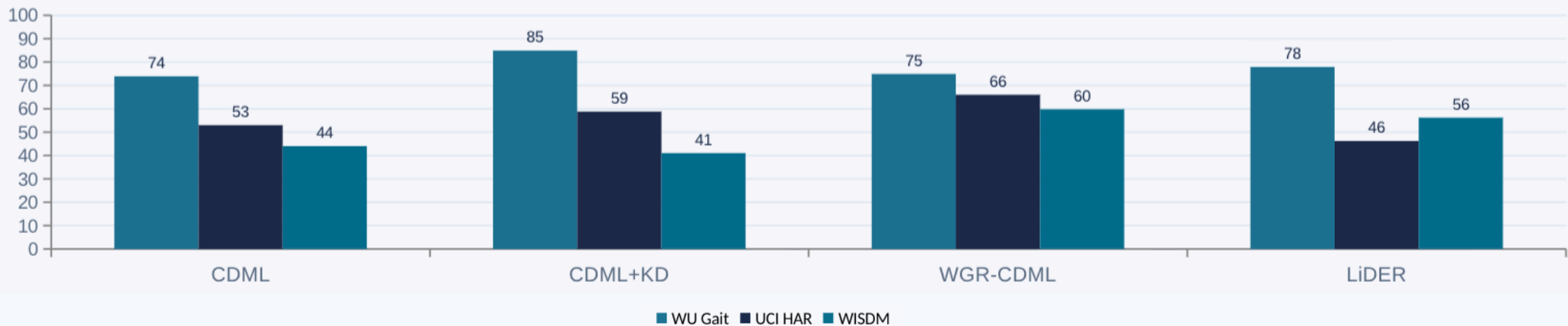
LiDER

CDML + Lipschitz Regularization

Bounds the embedding layer's spectral norm, plus a small 10% raw-replay buffer

$$\mathcal{L}_{\text{LiDER}} = \mathcal{L}_{\text{CE}} + w \cdot \max(0, \|\mathbf{W}_{\text{emb}}\|_{\sigma} - \kappa)^2$$

Average Accuracy by Method and Dataset (%)



Privacy & Backdoor Trade-offs (WU Gait)

Method	Accuracy	MIA (no-seed) AUC	IIA (no-seed) AUC	FSI acc.	Backdoor ASR
Std 15% (replay)	87.7%	56.9%	51.1%	90.3%	42.9%
CDML	73.9%	51.0%	48.7%	1.6%	3.2%
CDML+KD	84.9%	49.0%	51.0%	0.4%	36.1%
WGR-CDML	74.9%	51.0%	44.8%	2.2%	2.8%
LiDER	77.9%	50.5%	47.6%	3.2%	50.4%

Best overall balance: WGR-CDML — strong accuracy, no raw data retained and good privacy resilience against all the attacks

LiDER: Replay Buffer Leaks

High WU Gait accuracy (77.9%), but the 10% raw-replay buffer carries poisoned windows forward — worst backdoor ASR of any method (50.4%).

CDML+KD: Knowledge Distillation Leaks

Helps smaller datasets (UCI HAR), but the distillation buffer propagates a poisoned Task 3 forward — 36.1% ASR on WU Gait.

Frozen Backbone + Per-Task Adapters = Zero Forgetting

After Task 1, the CNN backbone and base weights are frozen forever. Each new task gets its own small rank-8 adapter pair for the embedding and classifier layers.



1. Freeze (after T1)

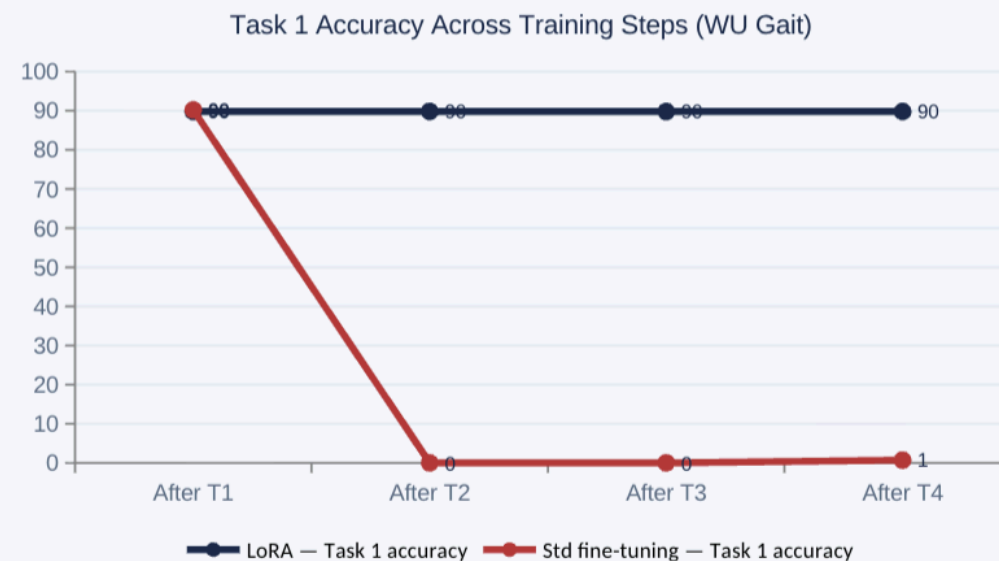
CNN backbone + base weights W_e, W_c locked forever

2. Adapt (per task) $y = Wx + \frac{\alpha}{r} B_k A_k x$

Rank-8 pair (A_k, B_k) trained for the embedding + classifier layers only

3. Restore (at inference)

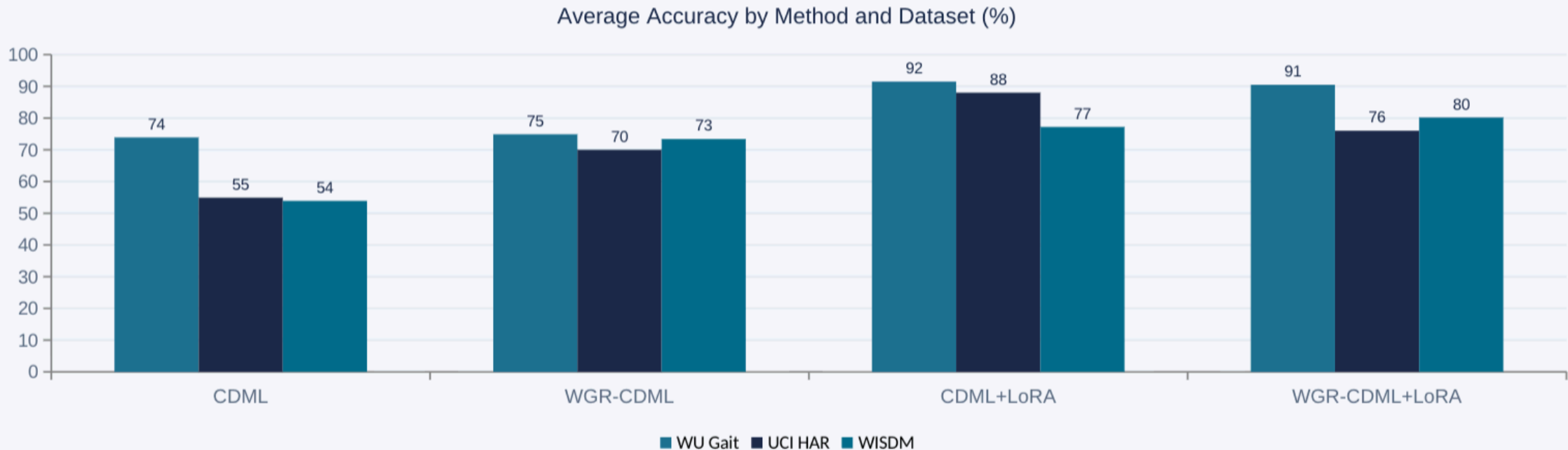
Load task k's saved adapter — full isolation, no cross-task interference



Zero forgetting is structural, not learned: Task 1's adapter is never modified after it is saved, so Task 1 accuracy is bit-for-bit identical at every later step.

CDML+LoRA: Zero Forgetting Plus High Privacy

Freezing the backbone (LoRA) and scrambling the embedding (CDML) solve forgetting and privacy independently. Combining them costs nothing on either axis.



CDML+LoRA is the first method to beat the 15% replay baseline (87.7% / 76.0% / 71.6%) on accuracy, on all three datasets, while keeping full privacy protection.

LoRA Alone Leaks Privacy — CDML Solves It

LoRA's frozen backbone still carries Task 1's signature, and does nothing to hide embedding geometry. CDML fixes both by scrambling the representation itself.

Model (WU Gait)	MIA AUC	IIA AUC	FSI acc.	Backdoor ASR
LoRA only (no-adapter)	41.2%	26.7%	86.0%	0.1%
CDML+LoRA (no-seed, no-adapter)	49.1%	56.5%	3.3%	1.7%

Task 1 is structurally Exposed on the LoRA only Model

Only Task 1 updates the shared backbone before it's frozen — its identity signature is preserved, while later tasks live in isolated per-task adapters. Since MIA and IIA both probe the shared representation, Task 1 is structurally the most exposed.

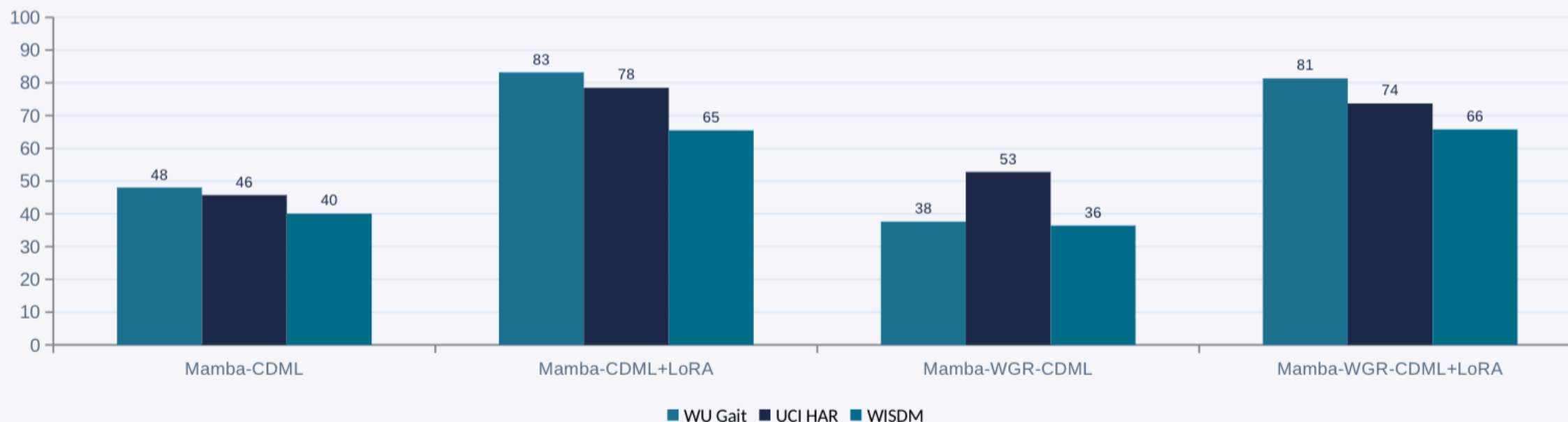
Small Datasets Still Need WGR

CDML+LoRA's backdoor ASR spikes to 32.6% on UCI HAR — its tiny per-task adapter overfits the trigger with so few samples. WGR-CDML+LoRA cuts this to 3.3% via the added wavelet-replay regularization.

Swapping the CNN for a Bidirectional State-Space Model

GaitMamba replaces the CNN with two parallel selective state-space branches, one reading the window forward, one backward, fused by a learned gate. CDML, LoRA and WGR all plug in exactly as before.

Average Accuracy by Strategy and Dataset (%)



Same pattern as GaitCNN: only the LoRA-based strategies reach zero backward transfer — the backbone choice doesn't change the anti-forgetting result.

Privacy Holds apart from the Backdoor

IIA and FSI follow the same access-control pattern seen throughout this thesis: near-chance without the seed, fully exposed with it. The backdoor attack gives much different results on this backbone.

Strategy (UCI HAR)	IIA (no-seed $\rightarrow$ oracle) AUC	FSI (no-seed $\rightarrow$ oracle) acc.	Backdoor ASR
Mamba-CDML	0.52 $\rightarrow$ 0.86	0.05 $\rightarrow$ 0.89	31.0%
Mamba-CDML+LoRA	0.40 $\rightarrow$ 0.99	0.06 $\rightarrow$ 0.93	40.0%
Mamba-WGR-CDML	0.41 $\rightarrow$ 0.95	0.08 $\rightarrow$ 0.96	38.0%
Mamba-WGR-CDML+LoRA	0.40 $\rightarrow$ 0.65	0.07 $\rightarrow$ 0.93	34.5%

MIA is not effective

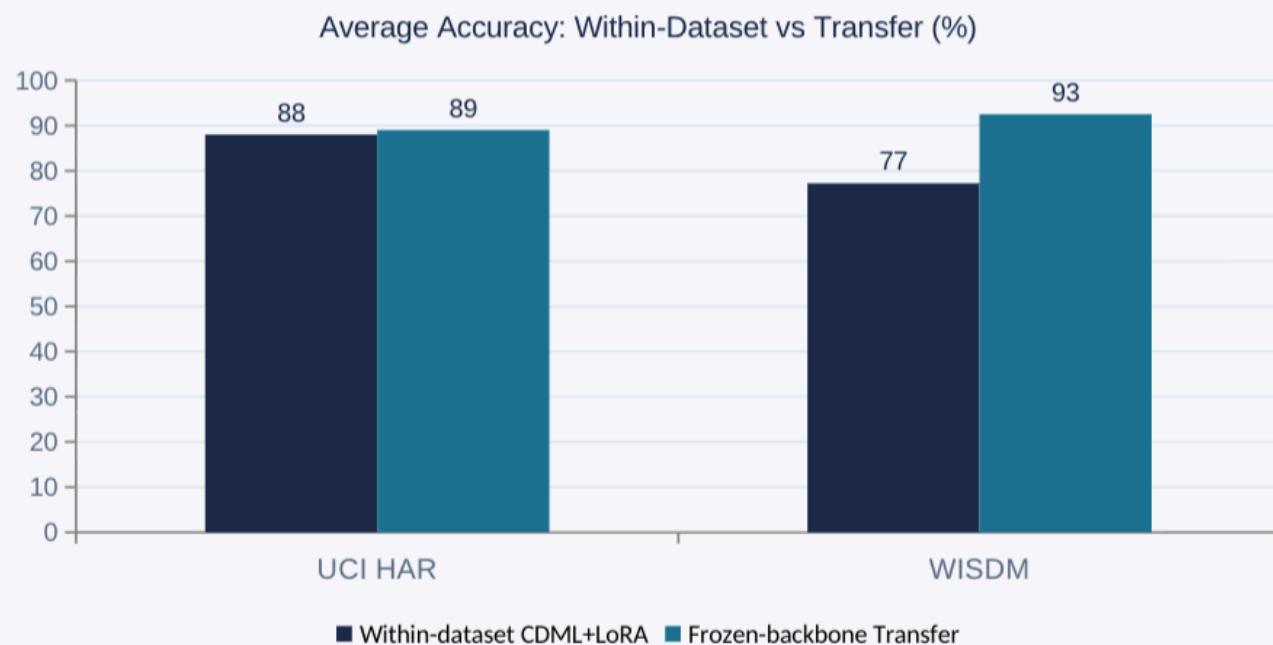
Membership inference sits at chance (0.49–0.52 AUC) even under full oracle access, no detectable per-sample memorization on this backbone, across every strategy and dataset.

Backdoor Defeats Every Defense

Unlike on GaitCNN, none of CDML, LoRA, or WGR meaningfully suppress the backdoor here — ASR stays 31–40% regardless of defense.

Pre-trained WU Gait Features Transfer

A GaitCNN backbone pre-trained on all 118 WU Gait subjects is frozen and reused as-is; only a LoRA embedding adapter and a CDML sequence are trained per task on the target dataset.



Why the Asymmetry?

UCI HAR keeps three walking variants (flat, upstairs, downstairs); the WU Gait backbone was trained on flat walking only. Its filters are a good, but imperfect, match for staircase dynamics — capping the transfer gain at +1.0 pp, versus +15.3 pp on WISDM's matching flat-walking data.

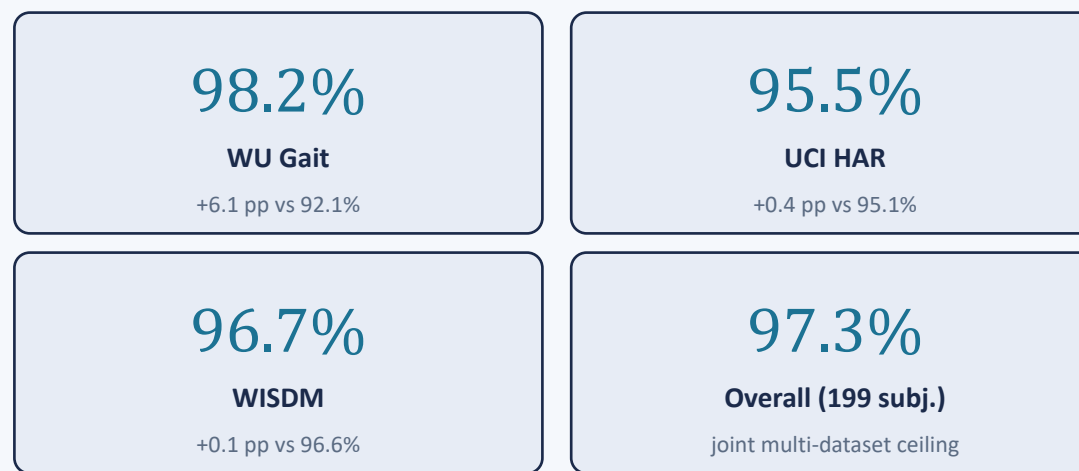
A frozen, pre-trained backbone is a cheap option when adding a new, smaller gait dataset to an already-deployed system.

Two More Angles: Warm-Starting and the Joint Dataset

A lighter form of transfer (initializing LoRA adapters from another dataset) accuracy

Experiment	UCI HAR	WISDM
Baseline (fresh init)	79.25%	91.88%
Order A (UCI → WISDM)	79.25%	92.24% ▲
Order B (WISDM → UCI)	78.40% ▼	91.88%

Joint Dataset (training on all subjects at once) accuracy



Warm-starting yields only marginal, asymmetric gains (± 0.4 – 0.9 pp) — joint training remains the upper threshold. Continual learning methods trail it most on UCI HAR (88.0% vs. 95.5%), leaving room for future work.

CDML+LoRA: The Best Overall Configuration

Dataset	Avg Accuracy	Forgetting	MIA AUC	IIA AUC	FSI acc.	Backdoor ASR
WU Gait	91.5%	0.0 pp	49.1%	56.5%	3.3%	1.7%
UCI HAR	88.0%	0.0 pp	49.9%	50.2%	3.9%	32.6%
WISDM	77.2%	0.0 pp	50.1%	52.1%	3.4%	0.0%

Alternative: WGR-CDML+LoRA — when backdoor robustness matters (cuts UCI HAR ASR to 3.3%) or WISDM is the primary target (80.2% vs 77.2%).

Catastrophic forgetting and privacy leakage can be solved together, without trading one off against the other.

Freezing the backbone (LoRA) and scrambling the embedding (CDML) are independent, composable mechanisms and their combination costs nothing on either axis.

Limitations & Future Directions

Limitations

Seed & adapter secrecy: every privacy guarantee collapses once the seed or adapters leak; secure distribution must be implemented.

Small datasets behave differently: task size interacts unpredictably with both forgetting and privacy mechanisms.

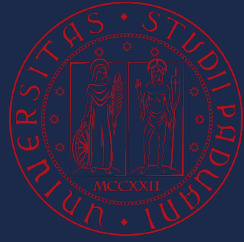
Future Directions

Larger, heterogeneous datasets: different sensor placements and a wider variety of activities.

Different Biometrics Domain: extending the evaluation to other biometric identification domains, such as face or voice recognition

Further Privacy Attacks: considering further privacy attacks, such as model inversion to assess whether CDML's modulation mechanism generalizes to other latent representations.

Together, these define a roadmap toward deployable, privacy-preserving continual biometric systems — built on the same core idea: forgetting and privacy protection do not have to compete, they have to coexist



UNIVERSITÀ
DEGLI STUDI
DI PADOVA



DIPARTIMENTO
MATEMATICA

Thank You!

Questions & Discussion

Tommaso Danieli · University of Padua — MSc. in Cybersecurity · Supervisor: Prof. Simone Milani · Co-Supervisor: Mattia Tamiazzo

github.com/Tommy270602/Gait_Continual_Learning